

What predicts survival in glioblastoma? A population-based study of changes in clinical management and outcome.

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Glioblastoma is the most common and most aggressive primary brain tumor in adults. Despite multimodal treatment, the median survival time is 15-16 months and 5-year survival rate 5%-10%. The primary goal of this study was to identify prognostic factors for survival in an unselected population of patients operated for glioblastoma. The secondary goal was to explore changes in outcome and the clinical management of this patient group over time. We identified 222 consecutive adults operated for glioblastoma between November 2012 and June 2016 at the Department of Neurosurgery, Sahlgrenska University Hospital in Gothenburg, serving a health care region in the western part of Sweden with 1.900.000 inhabitants. Clinical variables were identified and tested as predictors for prognosis in extended Poisson regression models. The results were compared with a previously published cohort from 2004 to 2008, before current standard of care based on molecular tumor diagnosis was fully implemented. Median overall survival was 1.07 years, which was significantly longer than in the 2004-2008 cohort (1.07 vs. 0.73 y, age- and sex adjusted HR = 1.89, $p < 0.0001$). Variables associated with longer survival in the multivariable model were MGMT promoter hypermethylation, non-central tumor location, complete resection of enhancing tumor, WHO performance status 0-1, unilateral tumor location, fewer lobes involved, younger age and no comorbidities. The median survival for patients with glioblastoma treated according to current standard treatment has moderately but significantly increased, with MGMT promoter hypermethylation as the strongest predictor for survival.

Survival Outcomes and Prognostic Factors in Glioblastoma.

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Background: IDH-wildtype glioblastoma is the most common malignant primary brain tumour in adults. As there is limited information on prognostic factors outside of clinical trials; thus, we conducted a retrospective study to characterise the glioblastoma population at our centre. Methods: Demographic, tumour molecular profiles, treatment, and survival data were collated for patients diagnosed with glioblastoma at our centre between July 2011 and December 2015. We used multivariate proportional hazard model associations with survival. Results: 490 patients were included; 60% had debulking surgery and 40% biopsy only. Subsequently, 56% had standard chemoradiotherapy, 25% had non-standard chemo/radio-therapy, and 19% had no further treatment. Overall survival was 9.2 months. In the multivariate analysis, longer survival was associated with debulking surgery vs. biopsy alone (14.9 vs. 8 months) (HR 0.54 [95% CI 0.41–0.70]), subsequent treatment after diagnosis (HR 0.12 [0.08–0.16]) (standard chemoradiotherapy [16.9 months] vs. non-standard regimens [9.2 months] vs. none [2.0 months]), tumour MGMT promotor methylation (HR 0.71 [0.58–0.87]), and younger age (hazard ratio vs. age < 50: 1.70 [1.26–2.30] for ages 50–59; 3.53 [2.65–4.70] for ages 60–69; 4.82 [3.54–6.56] for ages 70+). Conclusions: The median survival for patients with glioblastoma is less than a year. Younger age, debulking surgery, treatment with chemoradiotherapy, and MGMT promotor methylation are independently associated with longer survival.

Survival and prognostic factors for glioblastoma multiforme: Retrospective

single-institutional study.

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Glioblastoma multiforme (GBM) is the most common and aggressive primary brain tumor in adults. The standard management has been maximum surgical resection followed by adjuvant radiotherapy with concurrent chemotherapy followed by adjuvant chemotherapy. Although the survival rate of patients with GBM has improved with recent advancements in treatment, the prognosis remains generally poor. The median survival rates are in the range of 9-12 months and 2-year survival rates are in the range of 8%-12%. A single-institution retrospective review of 61 patients of GBM from 2012 to 2014. Data regarding patient factors, disease factors, and treatment factors were collected and survival has been calculated. A total of 61 patients with GBM were analyzed. GBM is commonly seen in sixth decade of life. Male to female ratio is 2.6:1. The right side of the brain is commonly involved with right frontal lobe being the most common site. The median follow-up was 4.6 months. The median survival of our patients was 8 months. The 1-year and 2-year survival rates were 20% and 3.27%, respectively. The overall survival and prognosis in patients with GBM remains poor despite of constant research and studies. Concurrent chemoradiotherapy followed by adjuvant chemotherapy with temozolomide should be used after maximal resection to improve the survival.

Treatment approach and survival from glioblastoma: results from a population-based retrospective cohort study from Western Norway.

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To evaluate treatment and survival from glioblastoma in a real-world setting. A population-based retrospective cohort study from Western Norway. 363 patients aged 18 years or older diagnosed with glioblastoma between 1 January 2007 and 31 December 2014. Overall survival and survival rates determined by Kaplan-Meier method, groups compared by log-rank test. Associations between clinical characteristics and treatment approach assessed by logistic regression. Associations between treatment approach and outcome analysed by Cox regression. Median overall survival was 10.2 months (95% CI 9.1 to 11.3). Resection was performed in 221 patients (60.9%), and was inversely associated with age over 70 years, higher comorbidity burden, deep-seated tumour localisation and multifocality. Median survival was 13.7 months (95% CI 12.1 to 15.4) in patients undergoing tumour resection, 8.3 months (95% CI 6.6 to 9.9) in patients undergoing biopsy and 4.5 months (95% CI 4.0 to 5.1) in patients where no surgical intervention was performed. Chemoradiotherapy according to the Stupp protocol was given to 157 patients (43%). Age over 70 years, higher comorbidity burden and cognitive impairment were associated with less intensive chemoradiotherapy. Median survival was 16.3 months (95% CI 14.1 to 18.5), 7.9 months (95% CI 6.7 to 9.0) and 2.0 months (95% CI 0.9 to 3.2) in patients treated according to the Stupp protocol, with less intensive chemoradiotherapy and with best supportive care, respectively. Surgical resection (HR 0.61 (95% CI 0.47 to 0.79)) and chemoradiotherapy according to the Stupp protocol (HR 0.09 (95% CI 0.06 to 0.15)) were strongly associated with favourable overall survival, when adjusted for clinical variables. In a real-world setting, less than half of the patients received full-course chemoradiotherapy, with a median survival comparable to results from clinical trials. Survival was considerably worse in patients receiving less intensive treatment. Our results point out a substantial risk of undertreating glioblastoma, especially in elderly patients.

Trends in glioblastoma: outcomes over time and type of intervention: a systematic evidence based analysis.

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Despite aggressive treatment with chemoradiotherapy and maximum surgical resection, survival in patients with glioblastoma (GBM) remains poor. Ongoing efforts are aiming to prolong the lifespan of these patients; however, disparities exist in reported survival values with lack of clear evidence that objectively examines GBM survival trends. We aim to describe the current status and advances in the survival of patients with GBM, by analyzing median overall survival through time and between treatment modalities. A systematic review was conducted according to PRISMA guidelines to identify articles of newly diagnosed glioblastoma from 1978 to 2018. Full-text glioblastoma papers with human subjects, ≥ 18 years old, and $n \geq 25$, were included for evaluation. The central tendency of median overall survival (MOS) was 13.5 months (2.3-29.6) and cumulative 5-year survival was 5.8% (0.01%-29.1%), with a significant difference in survival between studies that predate versus postdate the implementation of temozolomide and radiation, [12.5 (2.3-28) vs 15.6 (3.8-29.6) months, $P < 0.001$]. In clinical trials, bevacizumab [18.2 (10.6-23.0) months], tumor treating fields (TTF) [20.7 (20.5-20.9) months], and vaccines [19.2 (15.3-26.0) months] reported the highest central measure of median survival. Coadministration with radiotherapy and temozolomide provided a statistically significant increase in survival for patients suffering from glioblastoma. However, the natural history for GBM remains poor. Therapies including TTF pooled values of MOS and provide means of prolonging the survival of GBM patients.

Advances in Anti-Cancer Drug Development: Metformin as Anti-Angiogenic Supplemental Treatment for Glioblastoma.

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According to the WHO 2016 classification, glioblastoma is the most prevalent primary tumor in the adult central nervous system (CNS) and is categorized as grade IV. With an average lifespan of about 15 months from diagnosis, glioblastoma has a poor prognosis and presents a significant treatment challenge. Aberrant angiogenesis, which promotes tumor neovascularization and is a prospective target for molecular target treatment, is one of its unique and aggressive characteristics. Recently, the existence of glioma stem cells (GSCs) within the tumor, which are tolerant to chemotherapy and radiation, has been linked to the highly aggressive form of glioblastoma. Anti-angiogenic medications have not significantly improved overall survival (OS), despite various preclinical investigations and clinical trials demonstrating encouraging results. This suggests the need to discover new treatment options. Glioblastoma is one of the numerous cancers for which metformin, an anti-hyperglycemic medication belonging to the Biguanides family, is used as first-line therapy for type 2 diabetes mellitus (T2DM), and it has shown both in vitro and in vivo anti-tumoral activity. Based on these findings, the medication has been repurposed, which has shown the inhibition of many oncopromoter mechanisms and, as a result, identified the molecular pathways involved. Metformin inhibits cancer cell growth by blocking the LKB1/AMPK/mTOR/S6K1 pathway, leading to selective cell death in GSCs and inhibiting the proliferation of CD133+ cells. It has minimal impact on differentiated glioblastoma cells and normal human stem cells. The systematic retrieval of information was performed on PubMed. A total of 106 articles were found in a search on metformin for glioblastoma. Out of these six articles were Meta-analyses, Randomized Controlled Trials, clinical trials, and Systematic Reviews. The rest were

Literature review articles. These articles were from the years 2011 to 2024. Appropriate studies were isolated, and important information from each of them was understood and entered into a database from which the information was used in this article. The clinical trials on metformin use in the treatment of glioblastoma were searched on clinicaltrials.gov. In this article, we examine and evaluate metformin's possible anti-tumoral effects on glioblastoma, determining whether or not it may appropriately function as an anti-angiogenic substance and be safely added to the treatment and management of glioblastoma patients.

Conditional probability of survival and post-progression survival in patients with glioblastoma in the temozolomide treatment era.

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With standard treatment for glioblastoma (GBM) consisting of surgery followed by radiotherapy (RT) with concurrent and adjuvant temozolomide (TMZ), median survival is ~14.6 months. This is not as informative to patients who have survived for some time. Conditional probability of survival may offer more relevant survival estimates. Outcomes/conditional probability of survival and post-progression survival (PPS) estimates were retrospectively reviewed in the TMZ treatment era of 882 consecutive patients with a diagnosis of GBM from January 2004 to August 2010. Median age of entire cohort was 62 years including 62 % males. Baseline performance status (PS) was 0-1 in 67, 23 % had frontal lobe tumors, 58 % received concurrent RT/TMZ $\pm$ adjuvant TMZ. Survival (OS) was similar for those with frontal lobe tumors versus other locations ($P = 0.25$). OS for patients receiving standard RT/TMZ $\pm$ TMZ was 14.2 months. Age, PS, extent of surgery, therapy post-surgery had significant effects on OS. OS for entire cohort at 1, 2, 3, 4, 5 years was 43.4, 17.9, 10.4, 8.4, 7.2 % respectively. Conditional probability of survival of an additional year given survival to 1, 2, 3, 4, 5 years was 41.4, 58, 80.7, 85.7, 81.5 % respectively. Conditional probability of survival for those patients receiving concurrent RT/TMZ $\pm$ adjuvant TMZ was similar. Patients who progress >18 months after their initial treatment for GBM had significantly greater 2 and 5 year PPS as well as OS. Conditional probabilities of survival may provide more meaningful life expectancy predictions for survivors of GBM than conventional survival outcomes.

Leukopenia is a biomarker for effective temozolomide dosing and predicts overall survival of patients with glioblastoma.

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The median survival time of patients with glioblastoma is 14-16 months with a 5-year overall survival rate of 9.8%. Standard of care treatment includes radiation with concomitant temozolomide followed by cyclic temozolomide. If the patient develops myelosuppression (thrombocytopenia, leukopenia or anemia), the dose of temozolomide is reduced or stopped to avoid bleeding or infections. Recent studies have demonstrated that mild leukopenia is associated with increased overall survival in patients with glioblastoma. To confirm prior results showing that leukopenia is associated with increased overall survival as a primary outcome in patients with glioblastoma, the present study retrospectively collected complete blood counts from 141 patients with glioblastoma treated at the Beth Israel Deaconess Medical Center (Boston, USA) between January 2012 and December 2017. According to Kaplan-Meier analysis with a log-rank test, the presence of leukopenia was associated with increased overall survival ($P=0.008$).

Furthermore, patients with grade 2 leukopenia (Common Terminology Criteria for Adverse Events version 5.0) survived longer than those without myelosuppression ($P=0.024$). There was no difference in overall survival between patients with grade 1, 3 or 4 leukopenia and those without myelosuppression. Leukopenia was associated with longer survival independent of age or extent of surgery in Cox proportional hazards regression modeling ($P=0.00205$). A possible interpretation is that grade 2 leukopenia is a biomarker of adequate temozolomide dosing in a population with diverse DNA repair function, which may be the consequence of variable O6-methylguanine-DNA methyltransferase activity. A prospective dose escalation trial is necessary to determine if treatment-induced leukopenia is beneficial for all patients receiving temozolomide.

Glioblastoma survival in the United States before and during the temozolomide era.

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The standard-of-care treatment for newly diagnosed glioblastoma changed in 2005, when radiation therapy plus temozolomide chemotherapy replaced radiation therapy alone. It is not yet clear how this change in treatment has influenced patient survival in routine clinical practice, or if a survival benefit extends to patients older than those enrolled in the trial. Data from the Surveillance, Epidemiology, and End Results (SEER) Program was analyzed to compare survival of adult glioblastoma patients diagnosed from 2000-2003 to patients diagnosed from 2005-2008, in order to evaluate pre-temozolomide and post-temozolomide periods. The Kaplan-Meier method and Cox proportional hazards models were used. 6,673 patients with glioblastoma diagnosed from 2000-2003 and 7,259 patients diagnosed from 2005-2008 were identified. Median survival times of all patients diagnosed in the 2000-2003 and 2005-2008 periods were 8.1 and 9.7 months, respectively. Amongst patients treated with surgery and a radiation-containing regimen, median survival was 12.0 months in 2000-2003 and 14.2 months in 2005-2008. In the temozolomide era, median survival times ranged from a high of 31.9 months in patients age 20-29 to a low of 5.6 months in patients age 80 and older. The survival of patients with newly diagnosed glioblastoma improved from 2000-2003 to 2005-2008, likely due to temozolomide use. However, median survival time after glioblastoma diagnosis in the SEER population remains well under one year, largely driven by poor prognosis in elderly patients.

Clinical Effects of Immuno-Oncology Therapy on Glioblastoma Patients: A Systematic Review.

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The most prevalent and deadly primary malignant glioma in adults is glioblastoma (GBM), which has a median survival time of about 15 months. Despite the standard of care for glioblastoma, which includes gross total resection, high-dose radiation, and temozolomide chemotherapy, this tumor is still one of the most aggressive and difficult to treat. So, it is critical to find more potent therapies that can help glioblastoma patients have better clinical outcomes. Additionally, the prognosis for recurring malignant gliomas is poor, necessitating the need for innovative therapeutics. Immunotherapy is a rather new treatment for glioblastoma and its effects are not well studied when it is combined with standard chemoradiation therapy. We conducted this study to evaluate different glioblastoma immunotherapy approaches in terms of feasibility, efficacy, and safety. We conducted a computer-assisted literature

search of electronic databases for essays that are unique, involve either prospective or retrospective research, and are entirely written and published in English. We examined both observational data and randomized clinical trials. Eighteen studies met the criteria for inclusion. In conclusion, combining immunotherapy with radiochemotherapy and tumor removal is generally possible and safe, and rather effective in the prolongation of survival measures.
